

RESEARCH ARTICLE

Lightweight Truncated Fused-MirrorNet for Classification and Analysis of Histopathology Images

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ABSTRACT

Renal cell carcinoma is the primary cause of cancer-related mortality, highlighting the importance of early detection and accurate diagnosis. Manual histopathology image classification methods include limitations such as labor effort, time consumption, and interpathologist variations, which can lead to misdiagnosis, especially in the early stages. An autonomous solution based on deep learning is essential to overcome these constraints. However, vision-based models require significant processing resources and data sets, which pose difficulties for low-end infrastructures. In this study, we have described an approach for analyzing histopathology images using a lightweight truncated Fused-MirrorNet model. With its mirrored architecture, we use partial layer freezing and feature fusion approaches to improve performance. In kidney histopathology image analysis, our suggested strategy outperforms existing CNN and vision transformer models in histopathology image classification. We significantly reduced training time while preserving classification accuracy. The proposed model is deployable, scalable, and reproducible, allowing it to be used on low-end devices. Our strategy also makes it easier to create vision-based deep learning models by removing the requirement for sophisticated computational methodologies and procedures. The proposed model and the comparison models were trained and evaluated using histopathology images from two datasets. The experimental results reveal that the proposed model (Fused-MirrorNet) surpassed the performance of state-of-the-art models used for the classification of histopathology images. The proposed model achieves an accuracy of 92.60% and 90.00% in the TCGA kidney and BreakHis datasets, respectively. These findings indicate that the research conducted to develop the suggested model produced favorable outcomes.

1 | Introduction

Renal cell carcinoma (RCC) is a major cause of kidney cancer-related mortality worldwide. However, manual histopathology image diagnosis methods currently in use face challenges such as labor intensity, time consumption, and inter-pathologist variations, which can result in misdetection and misdiagnosis, particularly in the early stages. To overcome these limitations, developing an automatic method for histopathology image diagnosis is crucial. Conventional machine learning methods like support vector machine (SVM) and k-means, used for histopathology image analysis, often

require extensive feature engineering and struggle to capture complex patterns present in the images. For example, SVMs typically rely on handcrafted features, which require domain expertise and time-consuming manual effort to extract relevant information from the images. This process is not only labor-intensive but also prone to subjective biases and inter-observer variations. Additionally, SVMs can struggle with large-scale datasets and high-dimensional feature spaces, as they can be computationally expensive during both training and inference. Likewise, the traditional machine learning algorithm of k-means clustering has its limitations when it comes to capturing intricate patterns and variations present in

Research Highlights

- We propose Fused-MirrorNet, a lightweight deep-learning model for histopathology image classification.
- The model works on low-end systems while maintaining high accuracy and scalability.
- It achieved 92.60% accuracy on the TCGA kidney dataset and 90.00% on the BreakHis dataset.
- Fused-MirrorNet trains faster, requiring only 40 epochs to converge on both datasets.
- It is robust and generalizable, requiring minimal hyperparameter tuning for different datasets.

histopathology images. These limitations can impact the accuracy of the algorithm's output. Deep learning offers a solution by automatically extracting high-level abstract features from images, which eliminates the need for feature engineering. CNNs can capture complex features like spatial patterns and hierarchical representations effectively, enabling more accurate classification of histopathology images. The significance and effectiveness of deep learning for histopathology image classifications is supported by previous research in the field. Spanhol et al. (2016a), used a CNN-based deep learning method for breast cancer histopathology image analysis using patch-based training, showing that deep learning methods outperform machine learning models trained on the same dataset. Wang et al. (2021), used a model called TransPath, a transformer-based model trained using self-supervised learning for histopathology image analysis, which shows very promising results on histopathology image classification. The study carried out by Ding et al. (2021) and its result has demonstrated the potential of deep learning in histopathology image classification. The research by Alom et al. (2019) and Zhu et al. (2019) on CNN-based histopathology image analysis emphasizes the significance of deep learning techniques in cancer diagnosis. These studies provide valuable insights and inspiration for continuous research in this field. However, achieving reliable performance with deep learning models that rely on visual information poses challenges such as the high cost of computation and the need for robust and large datasets. Furthermore, the utilization of large vision-based models often necessitates extended training epochs and resource-intensive deployments. This can present challenges, especially for low-end infrastructures that may have limited computational resources.

To address these challenges, our study proposes a method for analyzing histopathology images using pre-trained and supervised deep convolutional neural networks (DCNN). In this study, we utilized the InceptionResNetV2 architecture introduced by Szegedy et al. (2016), which has shown impressive outcomes in various image classification applications. This architecture combines the strengths of both the Inception module and residual connections, resulting in improved representational power and feature extraction capabilities. The InceptionResNetV2 architecture has been widely adopted and demonstrated outstanding performance in different image analysis and classification tasks. Its effectiveness has been validated in medical image analysis as well.

For instance, Wang et al. (2021) applied the InceptionResNetV2 model in their research on histopathology image classification. The method introduced in Wang et al. (2021) achieved remarkable results in classifying histopathology images using this architecture. The use of InceptionResNetV2 in the proposed method for histopathology image classification leverages the model's ability to extract meaningful features and capture complex patterns within the images. By incorporating InceptionResNetV2 into our framework and optimizing its training approach, we achieved superior performance compared to existing methods. The method incorporates several modifications to the training approach to optimize performance and efficiency. These modifications include layer truncation, network replication, partial layer freezing, and feature fusion. Experimental results using the InceptionResNetV2 architecture as the backbone model for the proposed model called Fused-MirrorNet give better results for the classification of histopathology images compared to other existing methods. By fusing the extracted feature maps from both the network, one with layers kept trainable and the other with layers frozen and applying softmax classification, we achieve a significantly reduced model parameter size without compromising classification performance. This study presents a robust and fully automated deep learning framework for vision-based histopathology image classification, designed to address challenges related to reliable performance under computational and data constraints. The proposed approach leverages modified training strategies, pre-trained convolutional neural networks, and efficient feature fusion to achieve superior classification performance compared to existing methods. Owing to its rapid convergence and compact design, the framework is scalable, reproducible, and suitable for deployment on low-end devices, making it particularly valuable in resource-constrained clinical environments.

The contribution of this work is the introduction of an asymmetric dual-branch feature learning framework built upon a truncated InceptionResNetV2 backbone. Although both branches share an identical architecture, they are trained under different optimization constraints: one branch remains frozen to preserve stable low- and mid-level representations learned from ImageNet, while the second branch is fully trainable to adapt to domain-specific characteristics of histopathological images. The complementary features extracted from these branches are fused at an intermediate level, enabling the model to balance feature stability and adaptability—an aspect not typically addressed in conventional ensemble or mirror-based CNN approaches. Truncating the backbone at an intermediate depth preserves discriminative mid-level representations while reducing overfitting and computational overhead, which is particularly important for limited histopathological datasets. The asymmetric fusion of frozen and trainable branches enables the model to jointly leverage stable pretrained features and adaptively learned domain-specific patterns. The proposed framework is adaptable across multiple histopathology datasets and contributes toward the development of efficient and accurate AI-driven cancer detection systems.

1. In this study, a pre-trained InceptionResNetV2 architecture is truncated at an intermediate depth to reduce model size and computational complexity while retaining discriminative mid-level features. This truncated design enables faster end-to-end training without significant loss in classification performance.

2. This work proposes an asymmetric dual-branch framework based on a mirrored and truncated InceptionResNetV2 architecture. Although both branches share identical structures, they are trained under different optimization constraints. One branch is fully trainable and fine-tuned on different datasets to learn domain-specific representations for RCC classification, while the second branch remains frozen to preserve stable pretrained features.
3. The complementary representations extracted from these two branches are fused at an intermediate feature level using an efficient fusion strategy consisting of element-wise combination, global average pooling (GAP), and a lightweight classification head. This design allows the model to balance feature stability and adaptability, leading to a compact yet effective architecture for histopathological image analysis.

2 | Related Work

Histopathology image classification plays a crucial role in cancer diagnosis and prognosis. With advancements in deep learning techniques, researchers have explored various deep learning models for accurate and efficient histopathology image classification. Compared to traditional machine learning methods, such as SVM or random forests, CNNs have shown superior performance in histopathology image classification. The hierarchical nature of CNNs allows them to capture increasingly complex patterns as the information flows through the network, enabling accurate classification of histopathology images. Hence CNN-based models have been extensively employed for histopathology image analysis due to their ability to capture spatial dependencies in images. The model (Spanhol et al. 2016a; Krizhevsky et al. 2012; Huang et al. 2017; El-Nouby et al. 2021) utilized multiple convolutional layers and achieved better accuracy compared to traditional methods like machine learning. Recently, attention mechanisms have gained significant interest in deep learning due to their ability to selectively emphasize important features or regions within an input. For instance, models such as BreastNet, which are based on CNNs for breast histopathology image analysis and breast cancer diagnosis (Toğaçar et al. 2019), have achieved strong performance by incorporating the Convolutional Block Attention Module (CBAM) (Woo et al. 2018), which integrates both channel and spatial attention mechanisms. The channel attention module focuses on capturing interdependencies among channels by adaptively assigning weights based on their significance. This allows the network to prioritize relevant channels while suppressing less informative ones. On the other hand, the spatial attention module attends to specific spatial locations within an image, enabling the model to concentrate on relevant regions. CBAM enhances the network's discriminatory power by integrating channel and spatial attention, enabling it to capture more meaningful features and improve its classification performance. Transfer learning has emerged as a powerful technique for histopathology image classification. Wang et al. (2021) utilized transfer learning with a pre-trained CNN model to classify invasive breast carcinoma images. Their approach achieved state-of-the-art performance and demonstrated the efficacy of transfer learning in histopathology image analysis. They fine-tuned the pre-trained model on their specific dataset of histopathology images, enabling the model to learn task-specific features relevant to breast cancer classification. By leveraging the knowledge captured by the pre-trained model, their

approach achieved higher performance in classifying invasive breast carcinoma images (Tang et al. 2020; Gandomkar et al. 2017; Arevalo et al. 2016). Transfer learning offers several advantages in histopathology image analysis. Firstly, it reduces the need for large labeled datasets, which can be challenging to obtain in medical imaging tasks. Instead, it allows researchers to benefit from the knowledge already encoded in the pre-trained models. Secondly, transfer learning enables faster model convergence and reduces the overall training time. Generative Adversarial Networks (GAN) have also been investigated for classifying histopathology images. A GAN-based paradigm for classifying colorectal cancer was proposed by Zhang et al. (2019). Their model generated synthetic histopathology images, augmented by the existing training dataset to increase the number of samples in the training dataset and improve the classification performance. Multi-Instance Learning has also gained attention for histopathology image classification tasks where the diagnostic information may be present at different spatial levels. Li et al. (2020) proposed a MIL-based framework for lymphoma classification in histopathology images. Their model achieved promising results by considering multiple patches within the images, allowing for a more comprehensive analysis. Transformer models, which heavily rely on attention mechanisms, have also shown promise in histopathology image classification tasks. Vision Transformer (ViT) (Dosovitskiy et al. 2021; Chen et al. 2021; Abimouloud et al. 2025) is a notable example that shows the application of a Transformer to image classification tasks. ViT divides the image into patches, treats them as sequence inputs, and employs self-attention to capture global relationships among the patches. A recent study by Touvron et al. (2021) demonstrates the use of ViT for histopathology image classification. They demonstrated that ViT models achieved competitive results compared to CNNs on various histopathology datasets. By leveraging self-attention, ViT models effectively captured local and global dependencies, allowing them to classify histopathology images accurately. Table 1 summarizes research studies on deep learning techniques applied to biomedical image classification.

3 | Proposed Methodology

The model is built upon the InceptionResNetV2 architecture (Szegedy et al. 2016), which is the fundamental structure for effective feature extraction. To optimize the model's efficiency, transfer learning is utilized to decrease the number of parameters and maintain a powerful feature representation. Layer truncation and selective freezing are applied to adapt the model to the specific target classification task. By employing this strategy, the model selectively leverages the most relevant layers for the target task, thereby enhancing overall performance. Model concatenation and feature fusion methods are also employed to combine the learned features from multiple layers or models, enhancing the overall representation of the data. Transfer learning is an integral part of the model development process. By leveraging pre-trained weights from the InceptionResNetV2 model, the proposed model benefits from the knowledge learned from a large-scale dataset. This initialization enables the model to start with strong feature representations, facilitating faster convergence and improved performance. To refine the model further, fine-tuning is performed by adjusting the weights of the pre-trained model on the histopathological datasets. Partial layer freezing is applied to ensure that specific layers retain their pre-trained

TABLE 1 | Summary of existing deep learning approaches used in biomedical image classification.

References and dataset	Method	Performance metrics	Observation
Spanhol et al. (2016a)/BreakHis dataset	CNN (custom architecture)	Acc.: 85.6%–97.8% (binary), Acc.: 40.0%–82.0% (eight-class)	CNN achieved good results; performance varied across magnifications and decreased for multi-class classification.
Mahmood et al. (2025)/Microscopy breast image dataset	Dense convolutional network with attention mechanisms and multi-scale feature extraction.	Binary: Acc. 99.6%; eight-class: Acc. 99.4%	Strong diagnostic potential; limited comparative analysis with diverse external datasets.
Ding et al. (2021). PCam and NCT-CRC datasets	A Transformer-based model consists of both CNN and Transformer.	Acc.: up to ~97% (varies across magnification levels).	Lacks extensive benchmarking.
Alom et al. (2019)/BreaKHis dataset	IRRCNN (Inception Recurrent Residual CNN).	Acc.: 97.77% (binary), high sensitivity, specificity, <i>F</i> -score across magnifications	IRRCNN effectively distinguishes benign versus malignant tissues, residual connections stabilize training and enhance feature learning.
Zhu et al. (2019)/BreaKHis and BACH datasets	Assembling multiple compact CNNs (global + local branches with SEP block + model assembling).	BreaKHis: comparable with state-of-the-art; BACH: 87.5% patient-level (PL), 84.4% image-level (IL) accuracy	Hybrid model with channel pruning and ensemble improves generalization, Lack of feature-level interpretability.
Alabhya et al. (2021)/KMC and TCGA-LIHC datasets	LiverNet (deep CNN with CBAM + ASPP blocks for multi-scale feature extraction).	Acc.: 90.93%, high precision, recall, F1-score; efficient with 0.5739M parameters	Outperformed prior methods in HCC subtype classification; efficient design, but lacks feature-level interpretability.
Coudray et al. (2018)/TCGA NSCLC WSI dataset	Inception-V3 CNN (patch-based WSI analysis).	AUC = 0.97 (LUAD vs. LUSC), AUC: 0.73–0.86 (mutation prediction)	High classification performance is reported; however, mutation prediction is limited to select genes and lacks external clinical validation.
Spanhol et al. (2016b)/BreakHis dataset (7909 images, 82 patients)	Dataset introduction with baseline classifiers.	Acc.: 80%–85% (binary classification)	The work focuses on releasing a public dataset and evaluation protocol; reported results are preliminary and highlight the need for more advanced models.
Dosovitskiy et al. (2021)/ImageNet-1K, JFT-300M datasets	Vision Transformer (ViT) with 16 × 16 patch embeddings and standard Transformer encoder.	Acc.: 88.55% (ImageNet-1K, ViT-H/14), Top-1 accuracy	Demonstrates strong scalability with large-scale pretraining; however, performance degrades when trained on smaller datasets without extensive data augmentation.

(Continues)

TABLE 1 | (Continued)

References and dataset	Method	Performance metrics	Observation
Houssein et al. (2022)/MIAS and CBIS-DDSM mammography datasets	Optimized deep learning architecture using Improved Marine Predators Algorithm (IMPA).	Acc.: ~98%–99%	Metaheuristic optimization improves performance; however, evaluation is confined to mammography datasets without cross-dataset or external clinical validation.
Dhandapani et al. (2025)/Endometrial histopathology image dataset	Transfer Learning CNN (Xception-based) with Adam Gold Rush Optimization (AdGRO), including preprocessing and DConn-Net segmentation.	Acc.: 91.88%, TPR: 93.99%, TNR: 89.88%	High accuracy demonstrates effectiveness of AdGRO-optimized TL-CNN; however, evaluation is limited to a single dataset without cross-center validation.
Wan et al. (2026)/Wilms tumor whole-slide histopathology images	SGA-U-Net with hybrid attention for WT component segmentation and classification assistance.	Dice: 0.95 (blastema), 0.91 (epithelium), 0.88 (stroma)	High segmentation accuracy supports assistive diagnosis; however, validation is limited to WT and lacks multi-center external evaluation.
Kadirappa, Pujari Venkata, and Subbian (2025)/Private histopathology images (four classes, liver cancer)	Multi-Input Super-Resolution Neural Network (MISRNN) for image enhancement combined with lightweight classification networks.	PSNR: 39.12/33.98/31.02 dB (×2/×4/×6), SSIM: 0.948/0.868/0.807, Acc.: 96.7	Super-resolution preprocessing improves classification; however, evaluation is limited to a private dataset, and generalization to external datasets remains untested.
Kadirappa, Deivalakshmi, et al. (2025)/Kasturba Medical College (KMC), TCGA-LIHC, and LC25000 histopathology datasets	DeepHistoNet: attention-guided, densely connected residual and dilated convolutional CNN for multi-organ cancer classification.	Acc.: 97.1% (KMC liver), 99.8% (LC25000), AUC-ROC: 0.9867 (KMC liver).	Demonstrates high classification performance across multiple datasets.
Saremi and Kordbacheh (2025)/LC25000 colon histopathology dataset	Hybrid Graph-Transformer network (HG-TNet) combining transformer, CNN, capsule networks, and self-supervised rotation prediction for multi-scale feature extraction	AUC: 0.97–0.99 (multi-class ROC)	Evaluation is limited to LC25000, and cross-dataset generalization is untested.

(Continues)

TABLE 1 | (Continued)

References and dataset		Method	Performance metrics	Observation
Chaurasia et al. (2025)/Non-Melanoma Skin Cancer Segmentation (NMSCS) and Heidelberg WSIs (1.13M patches)		Multi-resolution Vision Transformer (ViT) incorporating 10×, 20×, 40×, 400× magnifications for six-class skin cancer subtype classification.	Quadratic weighted Cohen Kappa: 0.859–0.898 (internal), 0.791 (external), plus standard classification metrics.	Highlights interpretable attention maps. Generalizability beyond the external dataset remains to be further validated.
Zubair et al. (2025)/Gastric Histopathology Sub-size Image Database and HCRF dataset		Transfer learning framework with multi-channel attention mechanisms for interpretable gastric cancer classification.	Acc.: 99.07% (val), 98.48% (test) on Gastric dataset; Acc.: 99.84% (val), 99.65% (test) on HCRF	Relies on patch-level evaluation.
Abimouloud et al. (2025)/BreakHis dataset		TokenMixer hybrid CNN–ViT architecture inspired by ConvMixer and TokenLearner for efficient breast cancer subtype classification	Acc.: 97.02% (binary); 93.29% (multi-class); evaluated at 40×, 100×, 200×, 400× magnifications	The hybrid token-based design reduces computational complexity and improves accuracy; however, reliance on patch-level evaluation and limited validation on whole-slide images may restrict direct clinical applicability.
He et al. (2025)/TCGA colorectal cancer cohort (84 patients with histopathology, clinical, mRNA, and gene mutation data)		Multimodal deep learning framework using Multimodal Compact Bilinear Pooling (MCBP) to fuse histopathological images with clinical and molecular features for 5-year survival prediction.	AUC: 0.743 (image only); 0.771 (image + clinical); 0.773 (image + gene mutation)	Integration into real-world clinical workflows is not demonstrated.
Chanchal et al. (2025a)/Three RCC benchmark datasets		Multi-scale CNN with MCRT and GCDDL modules for RCC subtype classification.	Acc.: 91.67%, 97.14%, 97.24% for three different datasets, low FLOPs and parameters	Evaluated at patch level rather than full WSIs.
Khazaei Fadaeen and Rezaee (2023)/CRC-5000 and NCT-CRC-HE-100K datasets		Dilated ResNet + attention + DeepSVM ensemble for multi-tissue CRC classification.	Acc.: 98.75% (CRC-5000); 99.76% (NCT-CRC-HE-100K)	Clinical applicability not yet demonstrated.
Hameed et al. (2022)/Collected breast H&E dataset		Xception-based CNN for multiclass classification.	Acc.: 98% (original), 97.79% (Macenko); Sens.: 96%–99%; AUC-ROC: 0.998	High accuracy and robust features.

(Continues)

TABLE 1 | (Continued)

References and dataset	Method	Performance metrics	Observation
Chanchal et al. (2023)/KMC kidney histopathology dataset; BreakHis dataset	Efficient CNN-based RCCGNet with shared channel residual (SCR) block.	KMC: Acc. 90.14%; BreakHis: Acc. 90.09%	Evaluation limited to patch-level analysis
Prabhu et al. (2025)/Prostate Gleason and KMC Kidney datasets	ProGradNet CNN with CGSCR blocks for grading.	Acc.: 92.88% (Prostate); 92.68% (KMC)	High accuracy; WSI-level validation not done.

weights while allowing the remaining layers to be updated during training. This approach balances the preservation of useful learned features with the adaptation to the target task. The final model architecture incorporates all these techniques to create an effective framework for automated histopathology image diagnosis. By combining the strengths of the InceptionResNetV2 backbone (Szegedy et al. 2016), model replication, transfer learning, layer truncation and partial freezing, model concatenation, feature fusion, and fine-tuning within a unified network, the proposed model aims to achieve accurate and reliable diagnosis across multiple histopathological tissue types.

3.1 | InceptionResNetv2

The proposed Fused-MirrorNet model is built upon the training and fine-tuning of an InceptionResNetV2 backbone (Szegedy et al. 2016), leveraging transfer learning with ImageNet-pretrained weights. This involves truncating certain layers, replicating the truncated model, partially freezing layers in the replicated version, and fusing the features from both models. The InceptionResNetV2 model is created by fusing the two most widely used DCNNs, Inception and ResNet modules, and employing batch normalization instead of summations for the conventional layers. The additional modules enable a greater number of Inception blocks, resulting in a deeper network architecture. One of the primary challenges associated with very deep networks arises during training. This issue can be mitigated through residual connections, which facilitate effective gradient flow. By incorporating a large number of filters while maintaining reduced residual mappings, training stability and efficiency are significantly improved. The InceptionResNetV2 model utilizes a combination of multiple filter sizes and residual connections. This architectural design helps to address the degradation issue in deep networks and reduces training time. The network architecture is depicted in Figure 1, showcasing the arrangement of filters and residual connections. This design enhances the model's performance and facilitates effective feature learning during training. The Residual Inception Block is the fundamental unit of InceptionResNetV2. Before returning the filter bank back to the input after each block, a 1×1 convolution filter expansion layer in Equation (1) is used to make the filter bank large dimensional. The 164 layers of InceptionResNetV2 have an image input size of 224×224 pixels. The Residual Inception Block combines residual connections with convolutional filters of varying sizes. This design lessens training time and mitigates the degradation issue in deep networks by including these links. The refined InceptionResNetV2-based architecture for histopathological image analysis proposed in this study is illustrated in Figure 1. The variable used in Equations (1–4) has been defined in Table 2.

$$\text{Convolution}(I, K)_{ij} = \sum_{m=1}^M \sum_{n=1}^N I_{(i+m)(j+n)} \cdot K_{mn} \quad (1)$$

$$\text{MaxPooling}(I)_{ij} = \max_{m=1}^M \max_{n=1}^N I_{(i+m)(j+n)} \quad (2)$$

$$\text{Concatenation}(x_1, x_2, \dots, x_N) = [x_1, x_2, \dots, x_N] \quad (3)$$

$$\text{Batch Normalization}(X) = \frac{X - \mu}{\sqrt{\sigma^2 + \epsilon}} \odot \gamma + \beta \quad (4)$$

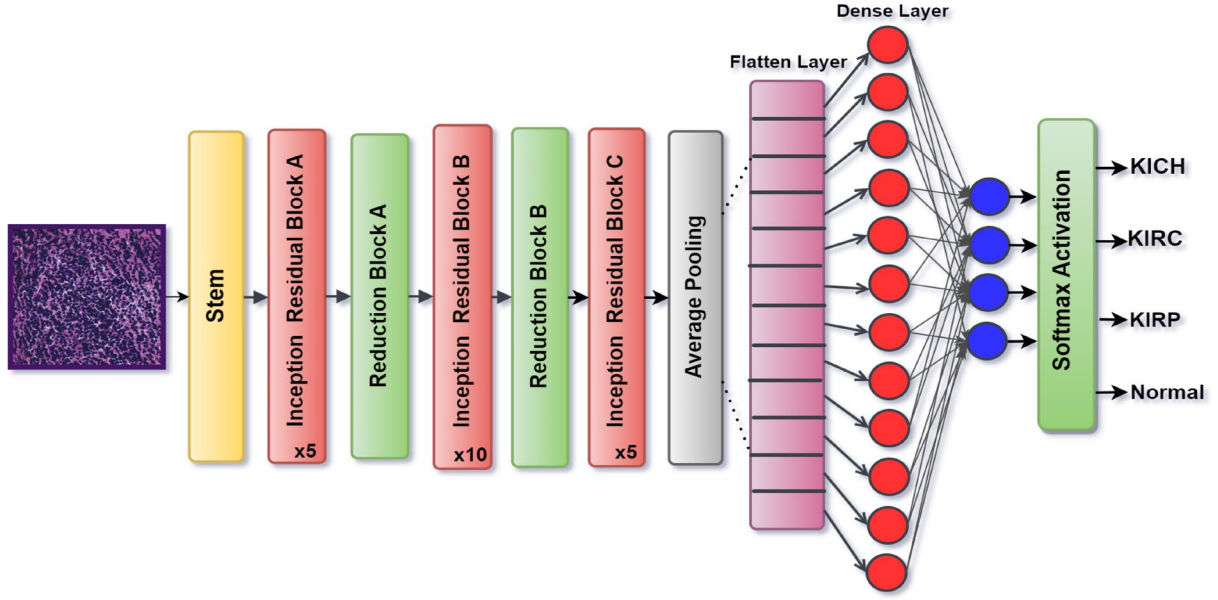


FIGURE 1 | Compressed architectural overview of the InceptionResNetV2 model.

TABLE 2 | Description of variables used in the mathematical formulations presented in this study.

Variable	Description
H	Image height
W	Image width
C	Number of feature maps
K	Kernel size
N	Batch size
i, j, m, n	Indices for spatial dimensions
x_{ij}	Value of pixel at position (i, j) in the input image
X_i	Input vector for the dense layer
W_i	Weights for the dense layer
B	Bias term for the dense layer
μ_j	Mean of feature values across the batch
σ_j^2	Variance of feature values across the batch
$\hat{x}_{ij}$	Normalized feature value at position (i, j)
ϵ	Small constant for numerical stability
$\sigma(\cdot)$	Activation function (e.g., sigmoid, ReLU)
p	Dropout probability
$x_1, x_2, x_3, \dots, x_n$	Input feature maps to be concatenated
y	Output feature map

3.2 | Feature Fusion Strategy

In the dual-branch architecture shown in Figure 2, feature fusion is performed at the feature-map level prior to any

pooling operation. Let $F_A \in \mathbb{R}^{H \times W \times C}$ and $F_B \in \mathbb{R}^{H \times W \times C}$ denote the output feature maps of the frozen and trainable truncated InceptionResNetV2 branches, respectively. Since both branches share the same truncation depth and input resolution, their feature maps are spatially and channel-wise aligned. Feature fusion is achieved using element-wise addition, defined in Equations (5)

$$F_{\text{fused}} = F_A + F_B. \quad (5)$$

This fusion strategy integrates stable pre-trained representations with adaptive domain-specific features while preserving spatial structure.

Subsequently, a GAP layer is applied to the fused feature map F_{fused} to produce a compact feature vector. This vector is then passed through a fully connected layer with ReLU activation, followed by dropout for regularization, and finally a softmax classifier for multi-class prediction. By performing fusion before pooling, the model retains richer spatial information, which is particularly beneficial for histopathological image analysis.

3.3 | Transfer Learning and Fine-Tuning of InceptionResNetV2 Model

This study utilized transfer learning, partial layer freezing, and fine-tuning approaches on the InceptionResNetV2 models to improve the performance of the limited training dataset consisting of histopathology images. On the other hand, transferring the same set of ImageNet features directly to both pipelines may lead to the generation of redundant features. To address this issue, a fine-tuning strategy was employed, in which selected layers of one of the mirrored InceptionResNetV2 networks were partially unfrozen to generate independent feature representations. During the transfer learning process, one InceptionResNetV2-Tiny network had its feature extraction layers frozen, as shown in Figure 3. This prevented the network from retraining those

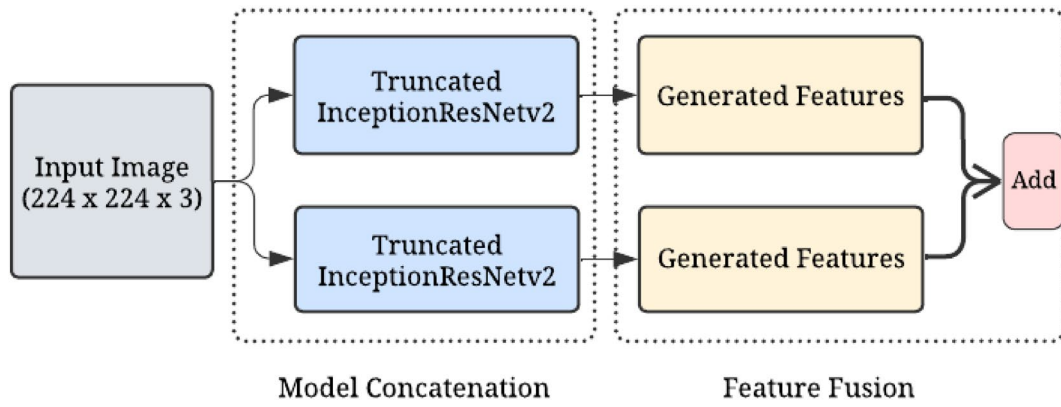


FIGURE 2 | Feature fusion and fine-tuning of truncated InceptionResNetV2 models.

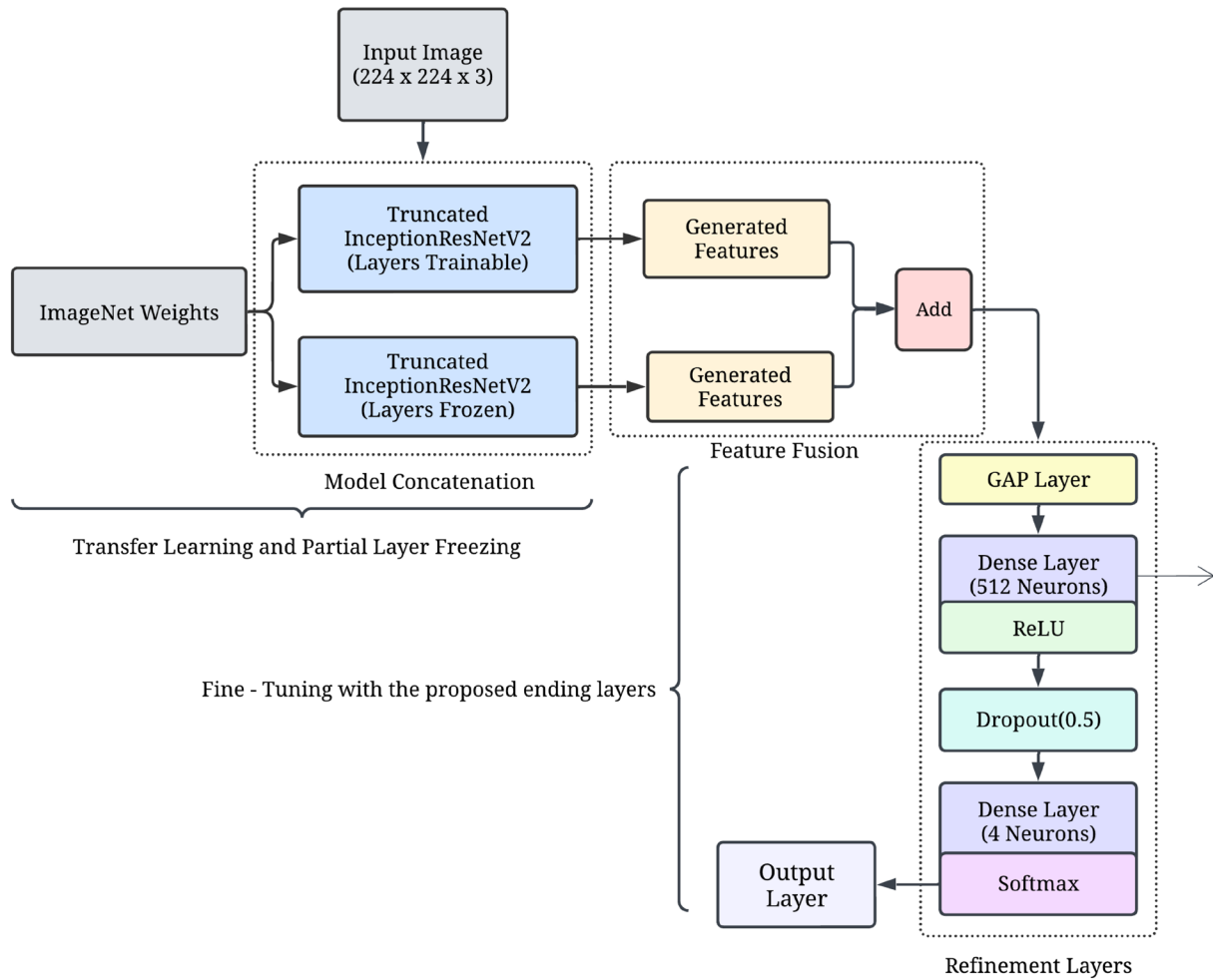


FIGURE 3 | Proposed model architecture.

layers, ensuring it preserved pretrained features while extracting representations from the histopathology image dataset. Using this approach, a mirrored version that has been given the name InceptionResNetV2-B was developed. This version unfroze all of the layers simultaneously and then retrained them on the histopathology dataset, that is, TCGA dataset, and BreakHis dataset to integrate the newly developed characteristics. On the other hand, InceptionResNetV2-A trained its bottom dense layers by utilizing the TCGA dataset and BreakHis dataset; at the same time, it kept the majority of the ImageNet features in its

upper layers. The proposed model successfully merged two comparable networks while maintaining separate feature sets, despite using fewer parameters. InceptionResNetV2-B generated a unique set of shared features by leveraging both the ImageNet and TCGA/BreakHis datasets, whereas InceptionResNetV2-A retained most of the pretrained ImageNet features and focused its training on the TCGA/BreakHis datasets for the lower dense layers. When fused into a single pipeline, the proposed model successfully recognized images. Its feature extraction capability was enhanced through feature map fusion, parameter

fine-tuning, and transfer learning. By combining two related networks, each with its own distinct feature set, the resulting model achieved a more robust feature representation while using fewer parameters.

3.4 | Proposed Model Architecture(Fused-Mirror NetV2)

As depicted in Figure 3, The input image is processed through two parallel architectures derived from the InceptionResNetV2 backbone. The residual connections in InceptionResNetV2 facilitate stable and efficient information flow across network layers, thereby mitigating performance degradation even as network depth increases. In the first branch, all layers are frozen, preventing weight updates during training and thereby preserving the pre-trained feature representations. In contrast, the second branch retains trainable layers, allowing the network to learn domain-specific features. In this study, the pre-trained InceptionResNetV2 backbone was truncated at an intermediate depth to reduce computational complexity while preserving discriminative feature representations. Specifically, truncation was applied at layer index 183 (counted from the output layer) with include_top=False, which corresponds to a mid-level feature extraction stage of the network. This depth was selected empirically through preliminary experiments evaluating multiple truncation points. Shallower truncation levels predominantly captured low-level texture features, whereas deeper layers exhibited increased computational cost and reduced generalization due to over-specialization toward ImageNet semantics. The selected truncation point provides a favorable balance between representational capacity and efficiency, making it well suited for histopathological image analysis. This truncation depth was consistently applied to both branches of the proposed dual-branch framework to ensure architectural symmetry. One branch was kept frozen to preserve stable pre-trained features, while the mirrored branch was fully trainable to adapt to domain-specific characteristics of kidney cancer images. The outputs of both branches were fused using element-wise addition, followed by GAP and classification layers. Instead of using the complete InceptionResNetV2 architecture, the network was truncated at layer index 183. The term “truncation point” refers to extracting a subset of the feature maps from the original design by eliminating certain layers. The output feature maps from both networks are combined element-wise and the resulting fused feature maps are then passed to a GAP layer, which reduces the spatial dimensions of the feature maps by computing the average value of each channel, thereby producing a compact feature representation. A dense layer is subsequently employed to model complex relationships among the extracted features. The ReLU activation function is applied to introduce non-linearity, enabling the network to learn more expressive feature representations. This is followed by an alpha dropout layer, whose mathematical formulation is presented in Equation (6), to enhance the robustness of the model. Dropout is a regularization technique that randomly sets a fraction of input units to zero during training, thereby reducing overfitting. Alpha dropout, a variant of this method, preserves the self-normalizing property of the network and

encourages the learning of more robust and diverse feature representations. Finally, the output of the dense layer, as described in Equation (7), is passed through a softmax activation function. The softmax layer produces a probability distribution over the target classes, corresponding to four classes for the TCGA dataset and eight classes for the BreakHis dataset. The class with the highest probability is selected as the final prediction. All architectural parameters, including the truncation depth, input resolution (224×224), and feature fusion strategy, are explicitly specified to ensure full reproducibility of the proposed model.

$$\text{Alpha_Dropout}(Y) = \begin{cases} \frac{X}{1-\alpha p}, & \text{with probability } 1-p \\ 0, & \text{with probability } p \end{cases} \quad (6)$$

$$\text{Dense_Layer}(Z) = \sigma \left(\sum_{i=1}^M X_i \cdot W_i + B \right) \quad (7)$$

4 | Training and Implementation

4.1 | The Cancer Genome Atlas (TCGA) Dataset

The kidney histopathology dataset used in this study comprises images from four categories: normal renal tissue, Renal Clear Cell Carcinoma (KIRC), Renal Papillary Cell Carcinoma (KIRP), and Chromophobe RCC (KICH). To ensure class balance, the number of Regions of Interest (ROIs) extracted from each class was standardized based on the class with the fewest samples. From the original 950 WSIs, 679 cases were selected where morphological and clinical features were highly representative. Patch extraction was performed under the supervision of expert pathologists, ensuring that only regions with clearly defined histopathological features were included. Patches were labeled as cancerous if the majority of their area contained abnormal tissue regions. In total, 4400 class-balanced patches were created, with each of the four classes contributing equally. The dataset was split slide-wise into training (85%) and testing (15%) sets, with a subset of the training data randomly selected for validation to prevent data leakage. Importantly, testing was performed on completely unseen WSIs to ensure robust evaluation. RCC is broadly categorized into several subtypes, among which the most clinically significant include clear cell renal cell carcinoma (KIRC), papillary renal cell carcinoma (KIRP), and chromophobe renal cell carcinoma (KICH). Each subtype exhibits distinct molecular characteristics and clinical behavior. Clear cell RCC (KIRC), also referred to as ccRCC, is the most prevalent subtype, accounting for approximately 70%–80% of all RCC cases. It is characterized by abnormal cells with clear cytoplasm resulting from lipid and glycogen accumulation. Papillary renal cell carcinoma (KIRP) is the second most common subtype, representing approximately 10%–15% of RCC cases, and is distinguished by papillary architectural patterns and frequent associations with MET gene alterations.

Chromophobe renal cell carcinoma (KICH) is a relatively rare subtype, comprising about 5% of RCC cases, and is characterized by large polygonal cells with pale cytoplasm and prominent cell membranes, often associated with mitochondrial DNA abnormalities. In this study, histopathological whole-slide images were obtained from the TCGA data portal. From these slides, ROIs of size 1024×1024 pixels were extracted at 40 $\times$ magnification. The curated dataset includes four classes: normal kidney tissue, KIRC, KIRP, and KICH. A total of 679 cases were used, comprising 577 training cases and 102 test cases. From these cases, 4400 ROIs were extracted, with 3768 ROIs used for training and 632 ROIs for testing. The detailed class-wise distribution of cases and extracted ROIs is summarized in Table 3. The dataset used in this study was collected from various sources, resulting in variations in data format. This inconsistency has the potential to impact the performance of DCNN models. The images were preprocessed prior to training and testing using the proposed deep learning architecture. These steps included scaling the images to a resolution of 224×224 and normalizing them to have a zero mean and unit variance. Additionally, each pixel value was scaled to a range of 0 to 1 by dividing by 255, ensuring that all pixels have comparable values. The sample images from the TCGA dataset have been depicted in Figure 4.

TABLE 3 | TCGA dataset class distribution at 40 $\times$ magnification.

Dataset	Total cases		Extracted ROIs (1024×1024)	
	Training	Test	Training	Test
Normal	164	29	942	158
KIRC	157	28	942	158
KIRP	149	26	942	158
KICH	107	19	942	158
Total	577	102	3768	632

4.2 | BreakHis Dataset

The BreakHis Dataset (Toğaçar et al. 2019), which is being used as a validation dataset for the model consists of data that falls into two primary categories: benign and malignant. Further subdivisions of the benign category include B-Adenosis, B-Fibroadenoma, B-Tubular_adenoma, and B-phyllodes_tumor. The malignant category gets further subclassified as M-Ductal_Carcinoma, M-Lobular_Carcinoma, M-mucinous_carcinoma, and M-papillary_carcinoma. Out of this total, 1742 images were taken from the dataset used for the training process. In addition,

TABLE 4 | BreakHis dataset at 40 $\times$ magnification.

Class	Original training patches	Original test patches	Augmented training images
Adenosis	100	16	298
Fibroadenoma	219	35	657
Phyllodes tumor	105	19	287
Tubular adenoma	128	23	380
Ductal carcinoma	752	112	2256
Lobular carcinoma	138	18	414
Mucinous carcinoma	174	31	522
Papillary carcinoma	126	19	378
Total	1742	273	5192

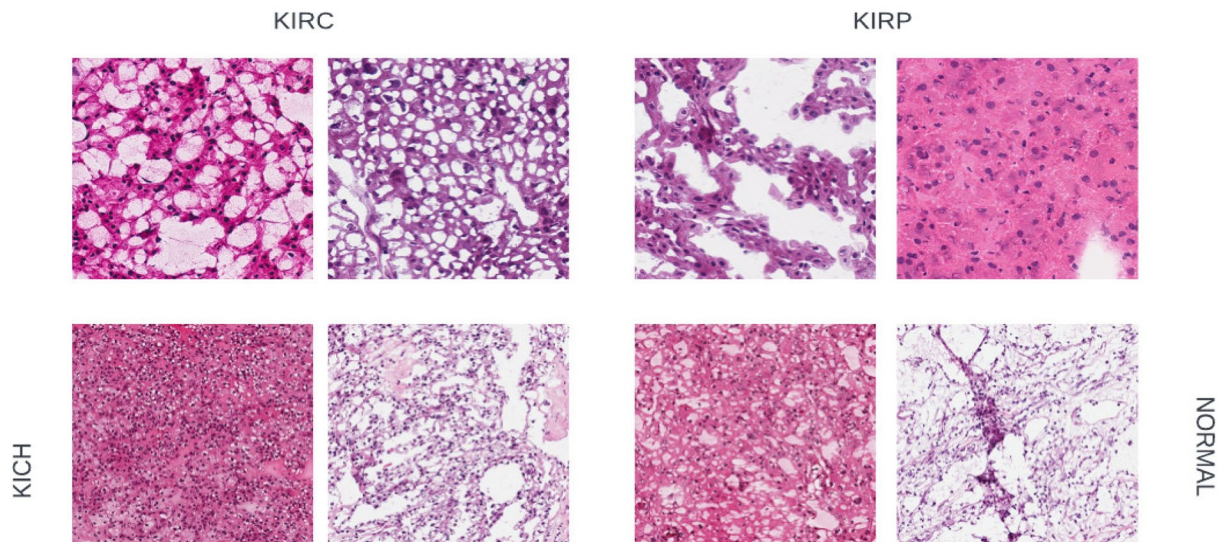


FIGURE 4 | Sample histopathology images from the TCGA dataset.

817 samples were reserved for validation purposes, which will allow the performance of the model to be fine-tuned and optimized. Lastly, a distinct subset of 273 images was set aside to verify the diagnostic accuracy of the model and test its efficacy on data that has not before been viewed. The dataset split has also been depicted in Table 4.

4.3 | Training Environment

During the training phase, the model was compiled with specific hyperparameters and a suitable loss function. A batch size of 16 was chosen to ensure efficient memory usage based on the dataset size and the available computational resources. The Adam optimizer with a learning rate (LR) of 0.001 was selected for its fast and adaptive training capabilities, allowing for quicker convergence within a relatively low number of epochs. To prevent the LR from being too high and potentially hindering the model's performance, a technique called ReduceLROnPlateau was employed. This technique automatically reduces the LR by a factor of 0.5 if the model's performance does not improve significantly after a certain number of epochs, helping to fine-tune the training process and improve accuracy without over-shooting. We updated the LR using the formula depicted in the Equation (8).

$$LR_{\text{new}} = LR \times 0.5 \quad (8)$$

The categorical cross-entropy loss function was used in the study. This loss function calculates the difference between the predicted and true probability distributions, encouraging the model to improve its accuracy by penalizing inaccurate predictions. By reducing categorical cross-entropy loss during training, the model learns to assign greater probabilities to correct class labels, improving its ability to classify inputs into distinct categories. The equation depicting the mathematics behind the same has been shown in Equation (9).

$$\text{Categorical Cross Entropy} = - \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(\hat{y}_{ij}) \quad (9)$$

In this formula, N : number of instances, M : number of classes, y_{ij} : true label of the i -th sample for the j -th class, and $\hat{y}_{ij}$: predicted probability of the i -th sample for the j -th class. The double summation is performed over all samples and classes.

The Adam optimizer (Kingma and Ba 2015) combines the advantages of RMSProp and stochastic gradient descent with momentum, two additional optimization techniques. Adam modifies the LR for each parameter separately, accounting for the average of the previous gradients and their squared gradients. Adam dynamically adjusts the LR during training by taking into account the first and second moments of the gradients, resulting in effective convergence and improved performance. This optimizer estimates the moments using exponential moving averages, which helps to stabilize and enhance the training process. Adam is a commonly used optimizer in the deep learning field because of its adaptability and efficiency, which has helped a variety of neural network topologies and applications

succeed. The mathematics behind this optimization technique has been depicted in the Equations (10–14). Where m_t is the average of previous gradients decaying exponentially. v_t is previously squared gradients decaying exponentially. g_t -current gradient, β_1 -rate of exponential decay for the momentary estimate. β_2 -the rate of exponential decay for the estimate of the second moment. t : current time step, η is LR. ϵ -small constant used to prevent division by zero error, θ_t -parameters before the update, θ_{t+1} -parameters after the update. $\hat{m}_t$ -estimation of the first moment without bias, $\hat{v}_t$ estimation of the second moment with bias correction. A list of hyper-parameters used in this study is shown in Table 5.

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t \quad (10)$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2 \quad (11)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (12)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (13)$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t \quad (14)$$

All experiments were conducted on a Dell G4-G3NSM workstation equipped with an NVIDIA Quadro P4000 GPU (8 GB VRAM) and 64GB system RAM. The proposed model and all reference models were implemented in Python 3 using TensorFlow 2.5.0 and the Keras deep learning framework.

5 | Results and Discussion

5.1 | Performance Evaluation Metrics

Performance metrics (Chanchal et al. 2024, 2021; Mahmood et al. 2025) are statistical measures used to find the accuracy of a model's predictions in relation to the actual or ground-truth labels. In this particular study, the model's performance is evaluated using the following set of metrics:

TABLE 5 | Software tools and hyperparameters setting.

Runtime environment	Python with version 3.8
Libraries	TensorFlow 2.12.0 and Keras 2.12.0
Computational-hardware	Google Colab using NVIDIA Tesla T4 GPU
Optimizer used	Adam with Initial learning rate of 0.0001
Loss-function	CCE (categorical cross entropy)
Activation-function used	ReLu and Softmax
Weight initializer	Kaiming uniform
Mini batch-size	16
No of training epochs	40

$$\begin{aligned}
\text{Recall} &= \frac{TP}{TP + FN} & \text{Precision} &= \frac{TP}{TP + FP} \\
\text{Accuracy} &= \frac{TP + TN}{TP + FP + TN + FN} \\
F1 \text{ score} &= \frac{2TP}{2TP + FN + FP} \\
\kappa &= \frac{p_o - p_e}{1 - p_e}
\end{aligned}$$

where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. In the computation of Cohen's Kappa, p_o represents the observed agreement between predicted and ground-truth labels, and p_e denotes the expected agreement occurring by chance.

5.2 | Visual Interpretability Analysis

To understand the decision-making process of the proposed dual-branch network, we performed a visual interpretability analysis using feature visualization and Grad-CAM (Chanchal et al. 2025b; Mahmood et al. 2025). These techniques help to reveal which regions of histopathological images contribute most to the model's predictions, providing insight into learned representations and ensuring clinical relevance. Attention maps were generated from the deep feature representations of the fused network before the final pooling layer. These maps highlight regions where the model focuses when extracting discriminative features for classification. For instance, in renal histopathology images, attention maps consistently emphasize abnormal tissue structures, tumor nuclei, or distinctive cytoplasmic patterns that are characteristic of KIRC, KIRP, or KICH subtypes. This confirms that the network captures clinically meaningful regions rather than relying on irrelevant background artifacts. Grad-CAM was applied to the last convolutional layers of both branches of the network to generate class-specific activation heatmaps. Given an input image I and a target class c , Grad-CAM computes the gradients of the score for class c with respect to feature maps of the selected convolutional layer. These gradients are then aggregated to produce a coarse localization map, highlighting regions that contribute most to the class prediction. Equations (15) and (16) describe the computation of the Grad-CAM heatmap.

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad (15)$$

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (16)$$

where A^k represents the k -th feature map, y^c is the score for class c , and α_k^c are the importance weights.

As shown in Figure 5, the Grad-CAM heatmaps overlaid on the original images clearly delineate the histopathological regions that drive the model's predictions. For example, areas with abnormal nuclei and irregular cytoplasm receive higher activation, indicating that the model accurately focuses on

clinically relevant patterns for renal cancer classification. The combined use of attention maps and Grad-CAM provides complementary perspectives on model behavior. Attention maps show which regions are generally emphasized during feature extraction, while Grad-CAM highlights class-specific discriminative regions. Together, they validate that the dual-branch fused network effectively captures both global tissue architecture and local tumor characteristics, thereby enhancing reliability for clinical applications.

This proposed framework has important biological and clinical implications for RCC diagnosis and workflow support in histopathology. By learning discriminative patterns from histopathological regions at 40× magnification, the model effectively captures both global tissue architecture and local cellular characteristics that are critical for differentiating RCC subtypes, such as clear cell, papillary, and chromophobe RCC, as well as normal renal tissue. These features correspond to clinically relevant morphological cues routinely used by pathologists, including cytoplasmic appearance, nuclear morphology, architectural growth patterns, and stromal characteristics. From a clinical perspective, the proposed system can serve as a decision-support tool for pathologists by highlighting diagnostically relevant regions through visual explanations (e.g., Grad-CAM heatmaps). Such visual cues can assist pathologists in rapidly identifying ROI, reducing oversight in large whole-slide images and improving diagnostic consistency, especially in cases with subtle or heterogeneous tumor patterns. This is particularly valuable in RCC, where subtype differentiation has direct implications for prognosis, treatment planning, and patient management. The system can also contribute to workflow efficiency by acting as a pre-screening or second-reader tool, prioritizing suspicious regions for detailed examination and reducing the manual workload associated with exhaustive slide review. In resource-limited settings or high-throughput pathology laboratories, this can help optimize time allocation while maintaining diagnostic quality.

In this study, the model was also evaluated on the BreaKHis dataset, demonstrating its effectiveness in breast cancer histopathology by accurately distinguishing benign and malignant tissue patterns across multiple magnification levels. These results indicate that the proposed approach is not limited to RCC alone and has the potential to generalize across different cancer types, thereby supporting its broader clinical applicability in digital pathology. Overall, the proposed approach demonstrates potential to enhance diagnostic accuracy, efficiency, and reliability in histopathology analysis.

5.3 | Experimental Results

The performance of the proposed Fused-MirrorNet was evaluated and compared with several competitive models, including InceptionResNetV2 (Szegedy et al. 2016), BHCNet (Jiang et al. 2019), Mahmood et al. (2025), ViT (Chaurasia et al. 2025), and RoCNN (Chanchal et al. 2025b), using the TCGA kidney dataset with four classes and the BreakHis dataset with eight classes. In this section, we evaluate the

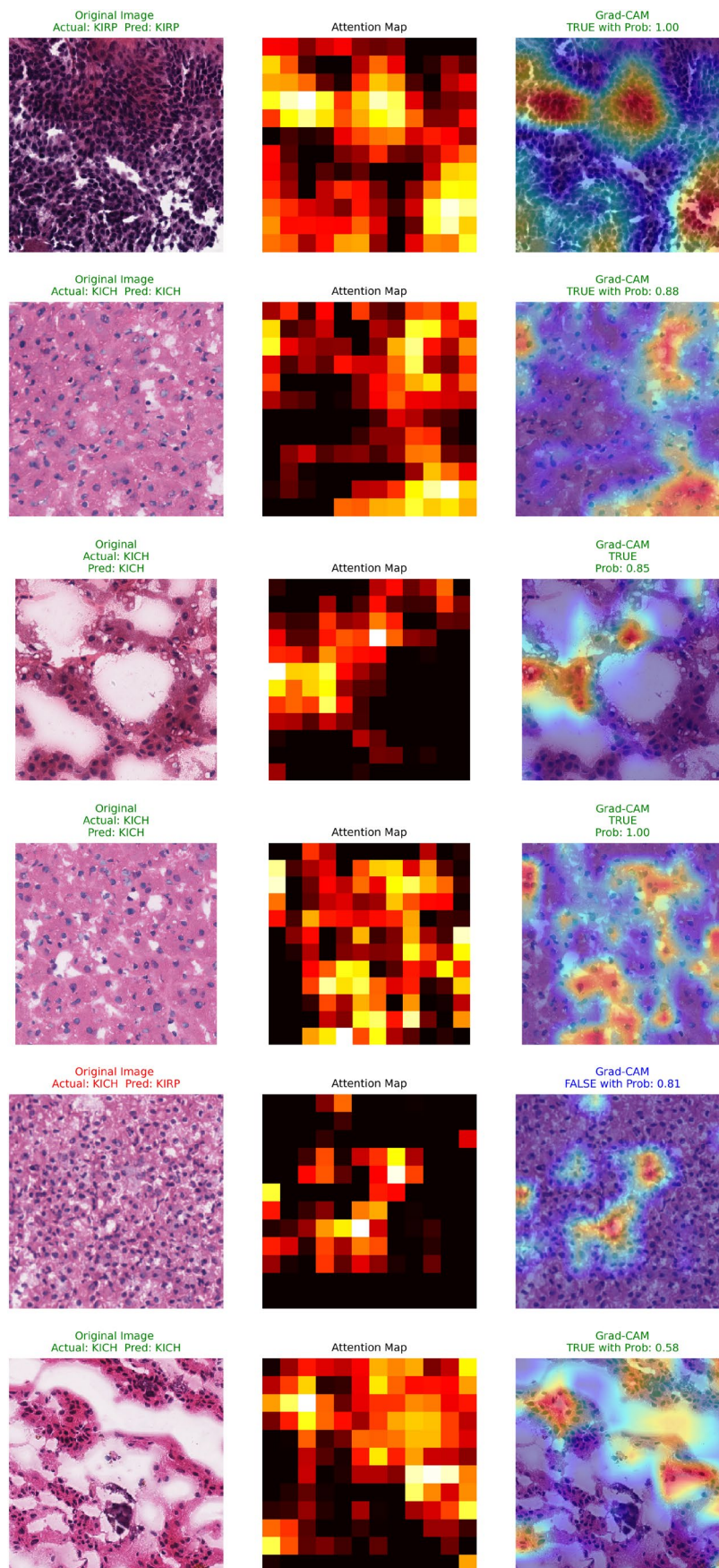
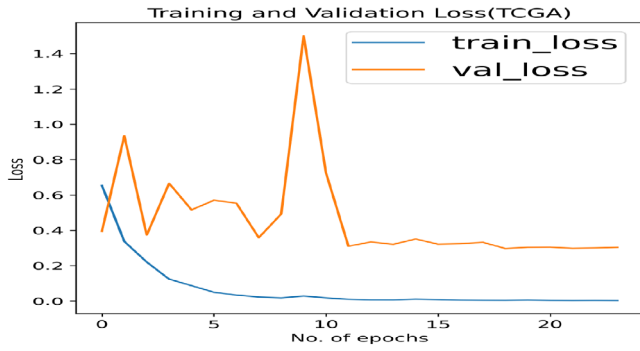
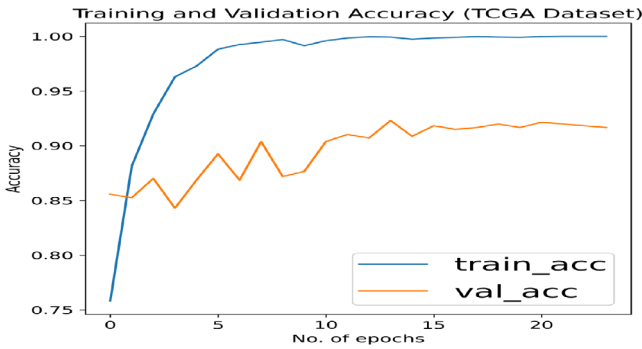


FIGURE 5 | Model interpretability on renal histopathology images using Grad-CAM. Left: Original images; Middle: Attention maps from deep feature representations; Right: Grad-CAM heatmaps highlighting regions contributing to the model's predictions.

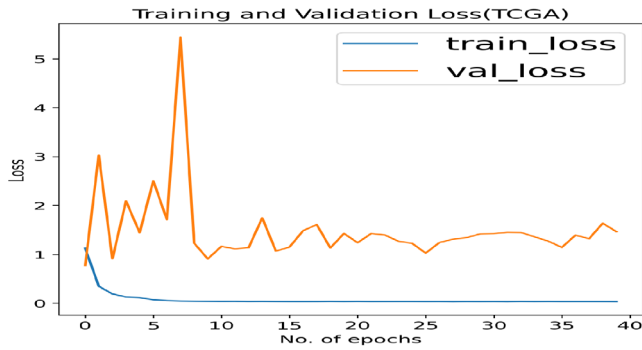


(a) Loss vs Epoch

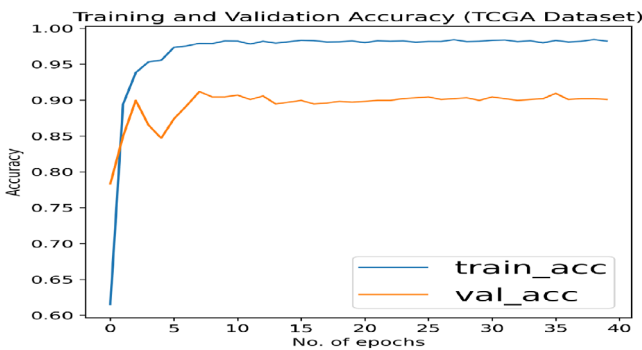


(b) Accuracy vs Epoch

FIGURE 6 | Plots of loss and accuracy for proposed model (fused-MirrorNet) on TCGA dataset.



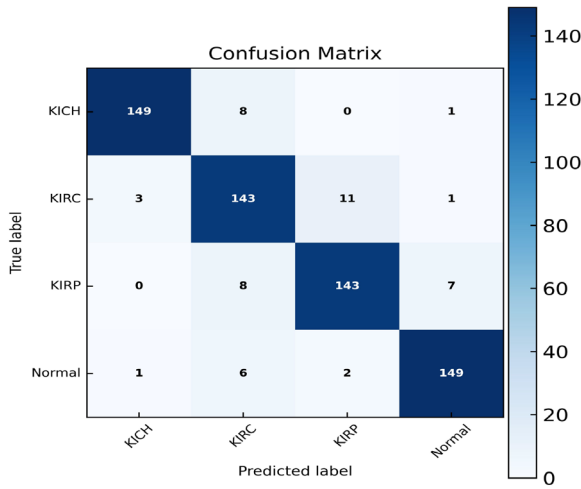
(a) Loss vs Epoch



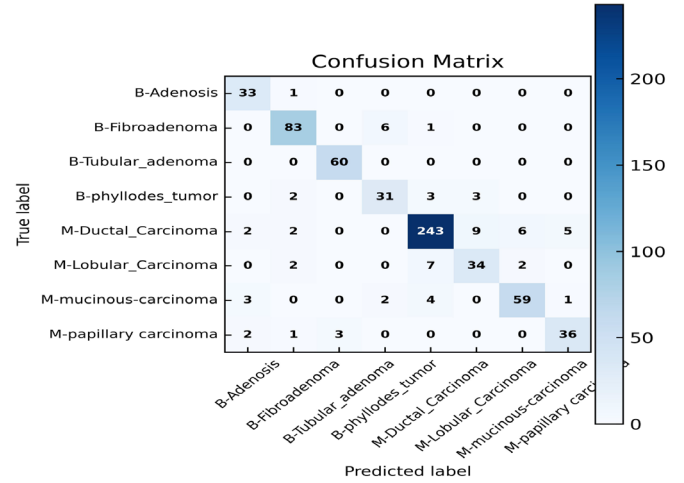
(b) Accuracy vs Epoch

FIGURE 7 | Plots of loss and accuracy for proposed model (fused-MirrorNet) on BreakHis dataset.

proposed Fused-MirrorNet model for accurately analyzing different forms of Kidney RCC as well as breast tumor histopathology using the TCGA and BreakHis datasets. The four classification categories are KIRC, KICH, KIRP, and Normal. Figure 6a,b shows the loss and accuracy curves for the TCGA dataset, while Figure 7a,b presents the loss and accuracy curves for the BreakHis dataset. A confusion matrix was used to evaluate the classification performance of the Fused-MirrorNet model on the validation data. Figure 8a shows the confusion matrix results for the TCGA kidney test dataset with four classes, while Figure 8b presents the confusion matrix results for the BreakHis test dataset with eight classes. The true label represents the ground-truth class of each image in the dataset, while the predicted label indicates the classification result for a given sample. The confusion matrix evaluates classification performance using True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values. A TP indicates that a sample is correctly classified as KIRC, KICH, KIRP, or Normal. A TN indicates that a sample is correctly identified as not belonging to any of these four classes. A FP occurs when a sample is incorrectly classified as one of the four classes, whereas a FN occurs when a sample belonging to one of the four classes is incorrectly classified. Fused-MirrorNet demonstrated class-specific precision, recall, accuracy, F1-score, and Cohen's kappa values. These evaluation metrics were used to compare the proposed model with existing methods. Fused-MirrorNet outperformed all competing models, achieving an accuracy of 92.60% on the TCGA dataset, as shown in Table 6, while the best accuracy reported by other models ranged from 86.70% to 90.30%. This performance difference indicates that Fused-MirrorNet effectively distinguishes different kidney RCC subtypes and normal tissue. The results demonstrate that Fused-MirrorNet consistently achieves higher performance with narrower 95% confidence intervals (CIs) across all evaluation metrics when compared with existing models. Narrower CIs indicate greater statistical stability and reliability, suggesting that the proposed model's performance is less sensitive to data variation. Across precision, recall, F1-score, accuracy, and Cohen's kappa, Fused-MirrorNet not only reports the highest mean values but also maintains limited CI overlap with competing methods, highlighting statistically meaningful improvements rather than marginal gains. In contrast, several reference models exhibit wider CIs, reflecting higher variability and less consistent performance across classes. For class-wise evaluation, Fused-MirrorNet achieves consistently high precision and recall with compact confidence bounds, indicating robust discrimination between kidney carcinoma subtypes and normal tissue. The improvements are especially notable in challenging classes, where competing models show larger uncertainty ranges. The proposed Fused-MirrorNet was further evaluated on the BreakHis dataset and compared with several competitive models, including InceptionResNetV2, BHCNet, Mahmood et al., ViT, and RoCNN, as summarized in Table 7. Among the existing approaches, the best accuracy achieved by the reference models ranged from 78.60% to 87.60%, with corresponding variations in precision, recall, and F1-score values. In contrast, Fused-MirrorNet achieved the highest overall performance on the BreakHis dataset, reporting an accuracy of 90.00% with a 95% CI of [0.875, 0.924]. The proposed model also attained superior precision (0.866), recall



(a) Confusion Matrices (TCGA dataset)



(b) Confusion Matrices (BreakHis dataset)

FIGURE 8 | Comparison of confusion matrices plots.

(0.874), and F1-score (0.865), all with comparatively narrower CIs, indicating stable and reliable classification performance. These results demonstrate that Fused-MirrorNet consistently outperforms existing models on the BreakHis dataset, highlighting its effectiveness and generalization capability for breast cancer histopathology image classification. Figure 9a,b presents the ROC curves for the TCGA and BreakHis datasets, respectively, while Figure 10a,b shows the corresponding Precision-Recall (P-R) curves. The results indicate that the proposed Fused-MirrorNet consistently outperforms other models on both datasets. The architecture achieves strong area-under-the-curve (AUC) performance, reflecting effective sensitivity and specificity. The ROC and AUC analyses confirm the model's reliable prediction capability across multiple classes, while the P-R curve analysis further demonstrates its effectiveness in handling class-wise diagnostic tasks. Overall, these evaluations validate the robustness of the proposed Fused-MirrorNet and highlight its potential for accurate and reliable histopathology image analysis, supporting its suitability for clinical applications.

5.4 | Computational Complexity Analysis

First, both branches are truncated at an intermediate depth of InceptionResNetV2, thereby removing the deepest and most computationally expensive residual blocks while retaining discriminative mid-level semantic features. Second, one branch is entirely frozen during training, which eliminates gradient computation and parameter updates for a substantial portion of the network, significantly reducing training-time complexity and memory overhead. Third, feature fusion is performed via element-wise addition followed by GAP, avoiding channel expansion and the heavy fully connected layers commonly used in ensemble architectures. As a result, the proposed model is considerably lighter than a full two-stream or ensemble InceptionResNetV2 configuration while maintaining strong representational capacity, making it suitable for

resource-constrained and clinical deployment scenarios. While on-device benchmarking on embedded platforms such as FPGA or NVIDIA Jetson is beyond the scope of this study, the reduced model depth, lightweight fusion strategy, and partial parameter freezing indicate that the proposed architecture is well suited for edge and low-resource deployment scenarios. Table 8 presents the computational complexity comparison of different variants of the proposed model.

5.5 | Limitations and Future Work

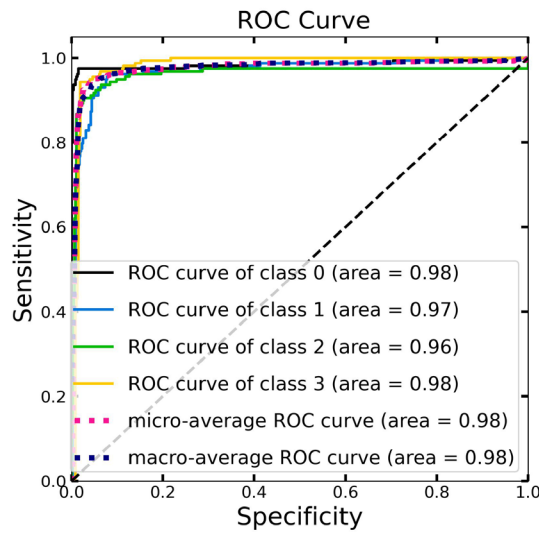
Despite promising results, the proposed model has several limitations, including its strong dependence on fully supervised annotations and the use of a limited, dataset-specific training set, which may restrict scalability and generalization. The model relies entirely on labeled histopathological data, and obtaining high-quality medical annotations is expensive and time-consuming, particularly for histopathology images where expert knowledge is required. Given the limited availability of annotated medical images, training deep learning models for histopathological image analysis remains a major challenge. Therefore, future work will consider substantial extension and diversification of datasets, as well as exploring alternative training strategies such as semi-supervised learning, active learning, and their synergistic integration. Leveraging both labeled and unlabeled data through techniques such as self-training, pseudo-labeling, and consistency regularization can improve model performance by exploiting the rich information present in unlabeled images. Additionally, incorporating active learning strategies to selectively annotate the most informative samples can reduce annotation effort while improving accuracy. Further investigations may focus on optimizing the balance between labeled and unlabeled data, developing effective data selection criteria, and designing novel algorithms that combine semi-supervised and active learning, thereby enhancing generalization, robustness, and clinical applicability.

TABLE 6 | Class-wise and overall performance comparison on the TCGA dataset with 95% confidence intervals (CI).

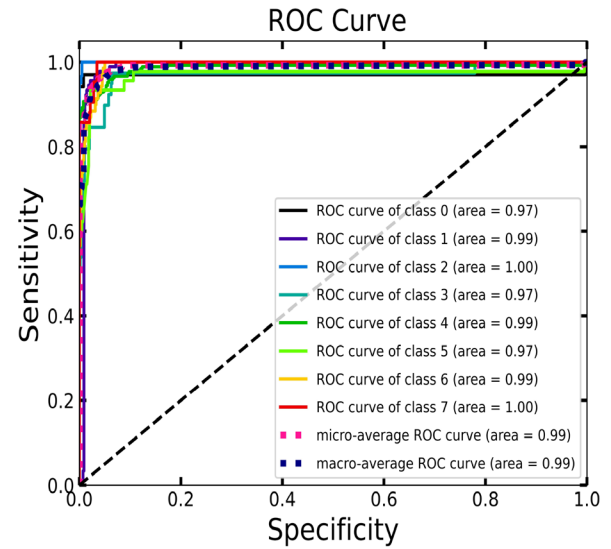
Metric	Class	IncRes (Szegedy et al. 2016)	BHCNet (Jiang et al. 2019)	Mahmood et al. (2025)	ViT (Chaurasia et al. 2025)	RoCNN (Chanchal et al. 2025b)	Proposed
Precision	Class 0	0.907 [0.858,0.951]	0.842 [0.784,0.896]	0.903 [0.853,0.947]	0.876 [0.825,0.922]	0.936 [0.897,0.970]	0.937 [0.894,0.975]
	Class 1	0.924 [0.876,0.965]	0.921 [0.876,0.960]	0.904 [0.861,0.949]	0.907 [0.860,0.950]	0.900 [0.854,0.944]	0.968 [0.938,0.993]
	Class 2	0.747 [0.689,0.804]	0.829 [0.769,0.889]	0.847 [0.786,0.903]	0.864 [0.803,0.918]	0.856 [0.797,0.909]	0.874 [0.824,0.921]
	Class 3	0.920 [0.876,0.958]	0.899 [0.848,0.944]	0.936 [0.896,0.970]	0.940 [0.900,0.974]	0.923 [0.880,0.963]	0.929 [0.886,0.969]
	Overall	0.875 [0.850,0.897]	0.873 [0.846,0.898]	0.898 [0.872,0.920]	0.897 [0.872,0.920]	0.904 [0.881,0.926]	0.927 [0.907,0.947]
Recall	Class 0	0.924 [0.878,0.963]	0.847 [0.786,0.902]	0.949 [0.913,0.981]	0.949 [0.912,0.981]	0.930 [0.887,0.969]	0.942 [0.903,0.975]
	Class 1	0.785 [0.721,0.849]	0.963 [0.931,0.988]	0.905 [0.856,0.946]	0.930 [0.884,0.968]	0.975 [0.947,0.994]	0.949 [0.915,0.980]
	Class 2	0.873 [0.816,0.924]	0.827 [0.765,0.885]	0.809 [0.744,0.870]	0.803 [0.742,0.862]	0.860 [0.805,0.912]	0.886 [0.833,0.936]
	Class 3	0.886 [0.835,0.934]	0.856 [0.797,0.912]	0.929 [0.890,0.969]	0.904 [0.860,0.947]	0.847 [0.789,0.904]	0.930 [0.888,0.969]
	Overall	0.867 [0.841,0.891]	0.873 [0.846,0.898]	0.898 [0.874,0.920]	0.896 [0.871,0.921]	0.903 [0.881,0.924]	0.926 [0.907,0.947]
F1-score	Class 0	0.915 [0.881,0.948]	0.844 [0.798,0.887]	0.925 [0.895,0.954]	0.911 [0.878,0.941]	0.933 [0.902,0.960]	0.939 [0.909,0.966]
	Class 1	0.848 [0.804,0.890]	0.941 [0.915,0.966]	0.904 [0.869,0.935]	0.918 [0.886,0.946]	0.936 [0.907,0.962]	0.958 [0.935,0.978]
	Class 2	0.805 [0.762,0.848]	0.828 [0.780,0.873]	0.827 [0.780,0.871]	0.832 [0.783,0.878]	0.858 [0.816,0.895]	0.880 [0.841,0.915]
	Class 3	0.903 [0.867,0.935]	0.876 [0.834,0.913]	0.932 [0.903,0.958]	0.921 [0.890,0.949]	0.883 [0.843,0.919]	0.929 [0.897,0.957]
	Overall	0.868 [0.842,0.892]	0.872 [0.845,0.897]	0.897 [0.873,0.920]	0.896 [0.871,0.920]	0.902 [0.880,0.924]	0.926 [0.906,0.947]
Accuracy	Class 0	0.957 [0.941,0.973]	0.922 [0.899,0.941]	0.962 [0.946,0.976]	0.954 [0.937,0.970]	0.967 [0.953,0.979]	0.970 [0.956,0.983]
	Class 1	0.930 [0.910,0.949]	0.970 [0.956,0.983]	0.952 [0.935,0.967]	0.959 [0.943,0.973]	0.967 [0.953,0.981]	0.979 [0.967,0.989]
	Class 2	0.894 [0.870,0.916]	0.914 [0.891,0.935]	0.916 [0.892,0.937]	0.919 [0.897,0.940]	0.929 [0.908,0.946]	0.940 [0.921,0.957]
	Class 3	0.952 [0.935,0.968]	0.940 [0.919,0.957]	0.967 [0.954,0.979]	0.962 [0.948,0.976]	0.944 [0.926,0.962]	0.965 [0.949,0.978]
	Overall	0.867 [0.840,0.891]	0.873 [0.846,0.899]	0.898 [0.873,0.921]	0.896 [0.872,0.921]	0.903 [0.880,0.924]	0.926 [0.907,0.946]
Kappa	Class 0	0.886 [0.842,0.929]	0.792 [0.735,0.847]	0.899 [0.859,0.938]	0.880 [0.837,0.920]	0.911 [0.872,0.946]	0.919 [0.879,0.954]
	Class 1	0.804 [0.748,0.857]	0.921 [0.886,0.954]	0.873 [0.825,0.913]	0.890 [0.848,0.928]	0.913 [0.875,0.949]	0.944 [0.913,0.971]
	Class 2	0.733 [0.678,0.788]	0.770 [0.709,0.828]	0.771 [0.710,0.826]	0.778 [0.717,0.837]	0.810 [0.754,0.860]	0.839 [0.789,0.886]
	Class 3	0.871 [0.825,0.913]	0.837 [0.783,0.884]	0.910 [0.874,0.944]	0.896 [0.857,0.933]	0.847 [0.794,0.893]	0.905 [0.864,0.943]
	Overall	0.822 [0.787,0.855]	0.830 [0.795,0.865]	0.864 [0.831,0.894]	0.862 [0.829,0.894]	0.871 [0.840,0.899]	0.902 [0.875,0.928]

TABLE 7 | Overall performance comparison on the BreKHis dataset with 95% confidence intervals (CI).

Metric	IncRes (Szegedy et al. 2016)	BHCNet (Jiang et al. 2019)	Mahmood et al. (2025)	ViT (Chaurasia et al. 2025)	RoCNN (Chanchal et al. 2025b)	Proposed
Precision	0.853 [0.818, 0.888]	0.756 [0.715, 0.798]	0.821 [0.780, 0.860]	0.876 [0.846, 0.905]	0.835 [0.795, 0.872]	0.866 [0.833, 0.898]
Recall	0.774 [0.734, 0.810]	0.740 [0.703, 0.780]	0.778 [0.743, 0.811]	0.812 [0.778, 0.848]	0.831 [0.797, 0.865]	0.874 [0.843, 0.906]
F1-score	0.805 [0.771, 0.839]	0.742 [0.703, 0.783]	0.790 [0.754, 0.824]	0.831 [0.796, 0.863]	0.826 [0.788, 0.862]	0.865 [0.833, 0.898]
Accuracy	0.838 [0.810, 0.867]	0.786 [0.754, 0.817]	0.838 [0.808, 0.864]	0.871 [0.845, 0.896]	0.876 [0.851, 0.902]	0.900 [0.875, 0.924]
Kappa	0.784 [0.747, 0.822]	0.720 [0.680, 0.761]	0.784 [0.747, 0.819]	0.830 [0.795, 0.863]	0.839 [0.806, 0.873]	0.870 [0.839, 0.901]

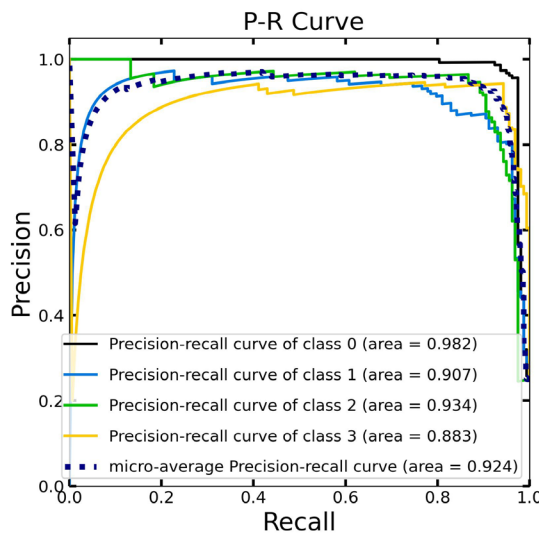


(a) ROC Plot (TCGA dataset)

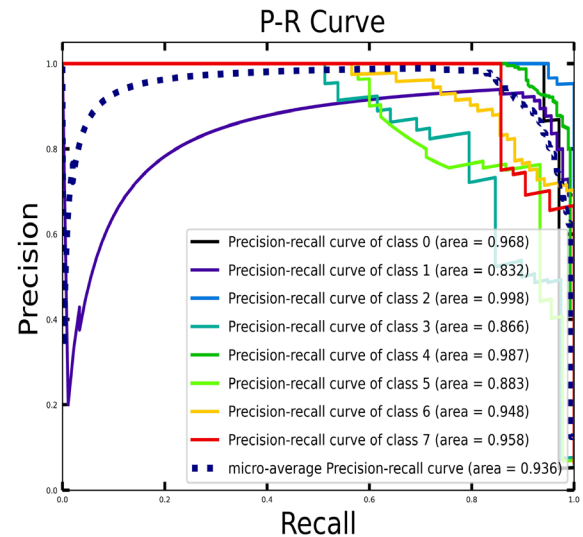


(b) ROC Plot (BreakHis dataset)

FIGURE 9 | Comparison of ROC plots.



(a) PR Plot (TCGA dataset)



(b) PR Plot (BreakHis dataset)

FIGURE 10 | Comparison of PR plots.

TABLE 8 | Comparison of model size and computational complexity.

Model configuration	Trainable parameters	Total parameters	FLOPs (approx.)
Full InceptionResNetV2 (baseline)	~55.9 M	~55.9 M	~13.2 GFLOPs
Two-stream InceptionResNetV2 ensemble	~111.8 M	~111.8 M	~26.4 GFLOPs
Proposed fused InceptionResNetV2	~18–22 M	~30–35 M	~6.0–7.5 GFLOPs

6 | Conclusion

In this study, we have proposed a deep learning framework, Fused-MirrorNet, for histopathology image classification, focusing on RCC subtypes and breast tumor classes. Unlike previous studies, this study proposes a mirrored and truncated InceptionResNetV2 that uses selective fine-tuning to extract both pretrained ImageNet features and domain-specific representations. The model is evaluated using standard metrics, including precision, recall, F1-score, accuracy, and Cohen's kappa, to measure its performance comprehensively. We have conducted the experiments on two well-established datasets, TCGA and BreakHis dataset to demonstrate its generalizability. Fused-MirrorNet achieved 92.60% accuracy on TCGA and 90.00% on BreakHis, outperforming other state-of-the-art models across all metrics. The model demonstrates fast convergence with reduced computational complexity, and Grad-CAM attention maps illustrate how it separates malignant tissue regions from the background, highlighting its effectiveness. Despite its promising performance, the model relies heavily on fully supervised annotations and a limited dataset, which could restrict scalability and generalization. Future work will focus on leveraging both labeled and unlabeled data through semi-supervised strategies to improve performance, reduce annotation effort, and enhance clinical applicability.

Author Contributions

Amit Kumar Chanchal: concept and design; drafting of the manuscript. **Shyam Lal:** supervision.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The source code, data, and trained model details are available at: https://github.com/chanchalnitk/Fused_MirrorNet.

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